LAB 19

Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report

Item	Detail
Topic	Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis
Pipeline stage	Monitoring and reporting
TSC reference	A8, A10
Duration	25 min
Folder	labs/Lab19-performance-report/
Files	Lab19-performance-report/report_stage.py, pipeline/report.py, Lab19-performance-report/solution/report.py
Mode	Runs ONLINE with your Alpaca paper keys in labs/.env, or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

After a year of paper trading, the investment committee wants a one-page tearsheet and a short report on the automated EMA 10/30 strategy: how the research backtest compares with the forward test, what it cost, whether any monitoring limit is breached, and whether the desk should allocate capital - produced automatically by the pipeline, every time.

Objective

Build `python -m pipeline report` to compare the research backtest with the runner's forward-test logs, run monitoring checks with thresholds, and generate a tearsheet and a markdown report with data-driven insights and a recommendation.

Before You Start

- Open a terminal in the labs folder and activate your virtual environment.
- Have your AI coding assistant open (for example Claude Code) in the labs folder if you want to rebuild the code from the prompts; the finished reference code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

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Step-by-Step Instructions

Step 1: Make sure the forward-test logs exist (A8)

The stage reads _outputs/Lab16_equity_log.csv and _outputs/trade_log.csv from the Lab 16 runner; if either is missing it runs the 2025 replay itself. Run python -m pipeline run --loop if you want fresh logs.

Check: _outputs/Lab16_equity_log.csv has 250 rows ending at USD 116,249.79.

Step 2: Build the backtest-vs-forward comparison (A8)

Ask your assistant for the first part of pipeline/report.py.

AI prompt - paste into your AI coding assistant
Create labs/pipeline/report.py. load_forward_test(cfg) reads the equity log and the BUY/SELL rows of trade_log.csv whose run_time falls in 2025-01-02 to 2025-12-31 (running pipeline.runner.replay quietly if a log is missing). In run(): backtest the configured strategy 2018-2025 with the config cost model and print a metrics table versus SPY buy and hold (total return, CAGR, volatility, Sharpe, Sortino, max drawdown, Calmar, win rate, profit factor, trades, exposure, alpha, beta) and the cost breakdown; backtest 2025 only (starting from 2024-12-31 so the first decision is warm) and compare it with the forward test: total return, Sharpe, max drawdown, orders and average fill slippage in bps versus the day's open; print the tracking difference and annualised tracking error. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: The comparison table has the columns 'Backtest 2025' and 'Forward test 2025 (paper)'.

Step 3: Add monitoring checks, the tearsheet and the report (A8, A10)

Ask your assistant for the monitoring section and the outputs.

AI prompt - paste into your AI coding assistant
In labs/pipeline/report.py add THRESHOLDS = {'tracking_abs': 0.02, 'slippage_bps': 5.0, 'drawdown_stop': 0.15} and monitoring checks with status OK or ALERT: tracking vs backtest, average fill slippage, drawdown stop rule (current and worst forward-test drawdown), kill switch (pipeline.risk.RiskManager) and diagnostics (read _outputs/Lab18_diagnostics.csv rows for the configured strategy: ALERT on any FAIL or if diagnose was not run for it). Save _outputs/Lab19_monitoring.csv; a 3 x 2 tearsheet Lab19_tearsheet.png (equity vs SPY with the forward-test year shaded, drawdowns, monthly-returns heatmap fixed at +/-8%, rolling 6-month Sharpe, backtest vs forward 2025, costs by type); and _outputs/Lab19_report.md with the tables (DataFrame.to_markdown), five data-supported insights and a recommendation: 'CONTINUE PAPER TRADING ...' if every check is OK, otherwise 'PAUSE AND INVESTIGATE: monitoring alerts on ...'. Paper results only - not investment advice. main(argv) for `python -m pipeline report`. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline report --help works.

Step 4: Generate the report (A8, A10)

Run: python -m pipeline diagnose (so diagnostics exist for ema_cross), then python -m pipeline report (same as: python3 Lab19-performance-report/report_stage.py).

Check: 2018-2025: strategy CAGR 9.02% vs SPY 14.15%, Sharpe 0.81 vs 0.78, max drawdown -14.19% vs -33.72%, beta 0.35, alpha +4.03%; costs USD 1,072 (drag 0.11% a year). 2025: backtest +17.04% (Sharpe 1.64) vs forward test +16.25% (Sharpe 1.65), 5 orders each, average slippage 1.0 bps, tracking difference -0.79%.

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Step 5: Read the monitoring checks (A8)

Read section [3]. Which limit is closest to being breached?

Check: Five checks OK: tracking -0.79% (limit +/-2%), slippage 1.0 bps (limit 5), drawdown current -1.34% / worst -4.85% (stop at -15%), kill switch off, diagnostics '1 WARN, no FAIL'.

Step 6: Trigger an alert on purpose (A8)

In pipeline/report.py change THRESHOLDS['tracking_abs'] to 0.005 and rerun python -m pipeline report. Then restore 0.02.

Check: The tracking check shows ALERT and the recommendation becomes 'PAUSE AND INVESTIGATE: monitoring alerts on tracking vs backtest'.

Step 7: Study the tearsheet (A10)

Open _outputs/Lab19_tearsheet.png: equity, drawdown, monthly-returns heatmap, rolling Sharpe, backtest vs forward, costs. Find the weakest months.

Check: The forward-test and backtest lines for 2025 almost overlap; the heatmap shows the weak months in red.

Step 8: Write the committee recommendation (A10)

Open _outputs/Lab19_report.md, read the generated insights and recommendation, and edit them in your own words, adding one risk the report does not mention.

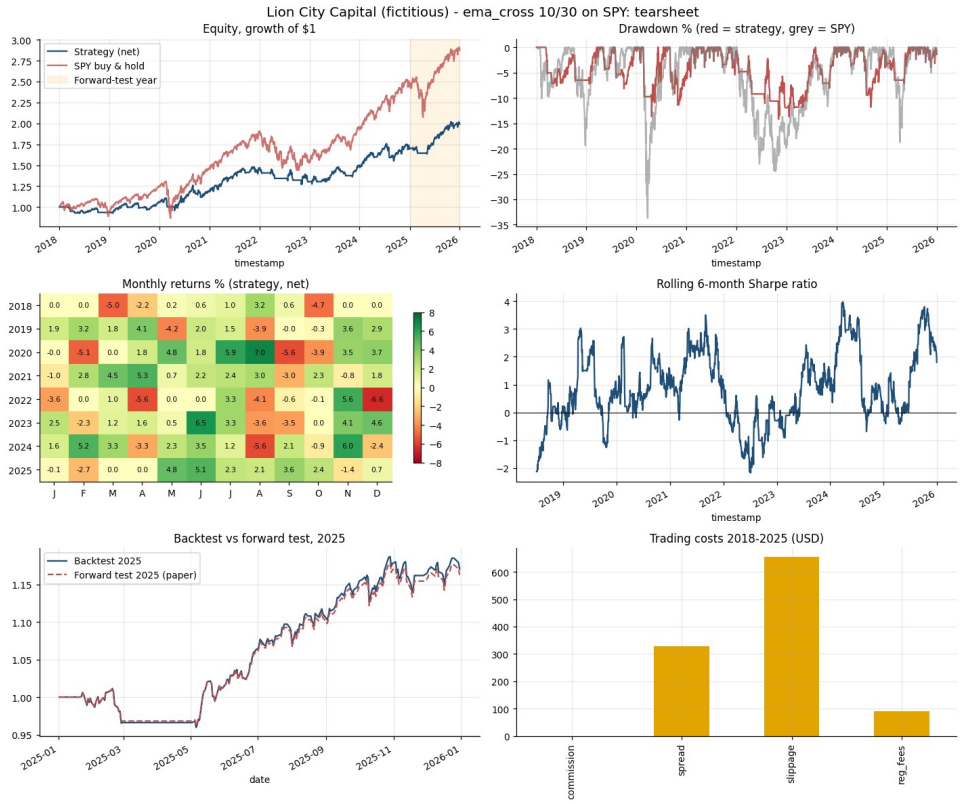
AI prompt - paste into your AI coding assistant
Read labs/_outputs/Lab19_report.md. Rewrite sections 5 and 6 for Lion City Capital's investment committee in plain English (under 200 words): what the data show, why the forward test matters, the honest message that the strategy reduces risk but has not beaten buy and hold, one additional risk (for example regime change or capacity), and a clear, conditional recommendation. Keep every number unchanged.

Check: A recommendation that continues paper trading with conditions (second signal, drawdown stop, risk-committee sign-off).

Expected Output

python -m pipeline report: 2018-2025 strategy CAGR 9.02% vs SPY 14.15%, Sharpe 0.81 vs 0.78, max drawdown -14.19% vs -33.72%, beta 0.35, costs USD 1,072 (drag 0.11% a year); 2025 backtest +17.04% (Sharpe 1.64) vs forward test +16.25% (Sharpe 1.65), 5 orders each, average slippage 1.0 bps, tracking -0.79%; 5/5 monitoring checks OK; recommendation: continue paper trading; files _outputs/Lab19_tearsheet.png, _outputs/Lab19_report.md and _outputs/Lab19_monitoring.csv.

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Lab 19 output (Lab19_tearsheet.png), offline mode on the bundled data.

Debrief

Comparing the backtest with the forward test is the final test of whether actions and outcomes match the model (A8). Automated monitoring with explicit thresholds turns that comparison into an alarm, and a good report turns numbers into trends, insights and a clear, conditional recommendation (A10) - including the honest message that the strategy reduces risk but has not beaten buy and hold.

Troubleshooting

Symptom	What to do
ImportError: tabulate	Run pip install tabulate (needed for the markdown tables).
Diagnostics check shows ALERT 'not run'	Run python -m pipeline diagnose for the configured strategy first.
Forward-test numbers differ	Rerun python -m pipeline run --loop with the default dates; --once rows outside 2025 are ignored.

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